

A Two-Dimensional Analogue of the Poincaré Conjecture

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Abstract

This paper studies the two-dimensional analogue of the Poincaré conjecture: that every closed, connected, and simply connected surface is homeomorphic to S^2 . We assume the classification theorem for closed surfaces as well as the Seifert–van Kampen theorem. We use these results to derive presentations of the fundamental groups of orientable and nonorientable surfaces. Finally, we compute the abelianizations of these fundamental groups and show that all nonspherical closed surfaces have a nontrivial fundamental group.

1 Introduction

I’ve always been interested in understanding the problems that mathematicians generally consider among the deepest and most difficult in the field. It provides me with a sense of grandeur when I think about how much there is left to learn and do within the field. That interest led me to read more about the Clay Millennium Prize Problems, where I learned that the Poincaré conjecture [Poi04] is the only one that has been solved. I know it’s a matter of debate what exactly “the most difficult problem in math” means, but I do think there’s a general consensus that the Clay problems are among the most difficult [NG06].

So, I decided to learn more about it and later found an article called *Manifold Destiny* in *The New Yorker*. I learned a lot about Perelman and the story behind the proof, and I was just amazed and filled with this childlike curiosity. I know the actual proof is far beyond the scope of this seminar, but I wanted to use this project as a step toward understanding the Poincaré conjecture in a setting where I can work through the math myself while learning some of the basic ideas of algebraic topology.

Theorem 1.1 (Two-dimensional Poincaré theorem). *Every closed, connected, simply connected surface is homeomorphic to S^2 .*

Our goal is to see how theorem 1.1 arises from studying concrete polygonal models and calculations of the fundamental groups. The path we take is

edge identifications $\longrightarrow$ generators and relations $\longrightarrow$ surface groups
 $\longrightarrow$ abelianization $\longrightarrow$ nontriviality.

There were a few things I found interesting while learning the material. For example, I never thought about how you could deform shapes into other shapes and how many polygonal models there are that can produce surfaces with particular fundamental groups. Even turning a square into a torus wasn’t something that was immediately apparent in my mind. It also took me some time to wrap my head around the fact that arcs in original polygons can only become loops after

their endpoints are identified in quotient space. Overall, it was an enjoyable process learning a new subfield of mathematics.

We have organized the paper as follows. Section 2 showcases assumed results and goes over terminology. In section 3, we illustrate how polygonal identifications become group relations using the torus and the real projective plane. Within Section 4 we prove the wedge-of-circles result and the two-cell attachment lemma. Section 5 derives the general surface-group presentations, while section 6 computes their abelianizations. Finally, section 7 proves theorem 1.1.

2 Background and assumed results

2.1 Surfaces and their classification

A *surface* is a topological space where every point has a neighborhood homeomorphic to an open subset of $\mathbb{R}^2$. It is considered *closed* if it is compact and has no boundary. On an orientable surface, one can choose a local orientation globally so that it remains consistent. For a nonorientable surface, there exists a closed path along which orientation is reversed.

A *genus* is the number of handles on an orientable closed surface. Σ_g is the orientable surface of genus g . For example, $\Sigma_0 = S^2$ and $\Sigma_1 = T^2$. We write N_k for the nonorientable surface obtained as the connected sum of k copies of the real projective plane $\mathbb{RP}^2$.

We assume the existence portion of the classification theorem for closed connected surfaces [Lee11, Chapter 6]:

Theorem 2.1 (Classification of closed surfaces, assumed). *Every closed, connected surface is homeomorphic to one of*

$$S^2, \quad \Sigma_g \ (g \geq 1), \quad N_k \ (k \geq 1).$$

2.2 Loops and fundamental groups

Let X be a topological space with basepoint x_0 . A loop based at x_0 is a continuous map $\gamma : [0, 1] \rightarrow X$ satisfying $\gamma(0) = \gamma(1) = x_0$. If a loop can be continuously deformed into another loop while keeping x_0 fixed, we say they represent the same element of the fundamental group. The homotopy class of a loop γ is denoted by $[\gamma]$. The fundamental group $\pi_1(X, x_0)$ is the set of all such classes.

The group operation is to first traverse one loop and then the other. This is also called concatenation. The constant loop is the identity. The inverse of a loop is the same path traversed backward. A space is *simply connected* if it is path-connected and its fundamental group is trivial.

We assume the Seifert–van Kampen theorem and the standard computation

$$\pi_1(S^1) \cong \mathbb{Z}.$$

For the version of Seifert–van Kampen used below, see, for example, [Hat02, Section 1.2].

2.3 Generators, relations, and abelianization

A presentation

$$\langle x_1, \dots, x_n \mid r_1 = 1, \dots, r_m = 1 \rangle$$

is defined as beginning with the free group on the generators $x_1, \dots, x_n$ and imposing the listed relations. The normal closure is the smallest normal subgroup containing the relators.

For group elements g and h , their commutator is

$$[g, h] = ghg^{-1}h^{-1}.$$

The abelianization of a group G is

$$G^{\text{ab}} = G/[G, G],$$

where $[G, G]$ is the subgroup generated by all commutators. Thus abelianization forces all elements to commute. We will consistently use the following implication going forward:

$$G^{\text{ab}} \neq \{0\} \implies G \neq \{1\}.$$

Meaning that a trivial group cannot have a nontrivial quotient.

This is the contrapositive to the logical statement that: if a group is trivial then every quotient of the group is trivial. Thus, we do not need to solve every group, just find a nontrivial quotient.

3 Polygonal models: two motivating examples

We call a polygon together with instructions for pairing its boundary edges a polygonal model. The final surface is the quotient we obtain after using the prescribed pairings.

3.1 The torus

We use a square and identify the opposite sides going in the same direction to make a torus. If we read the boundary counterclockwise we get the word

$$aba^{-1}b^{-1}.$$

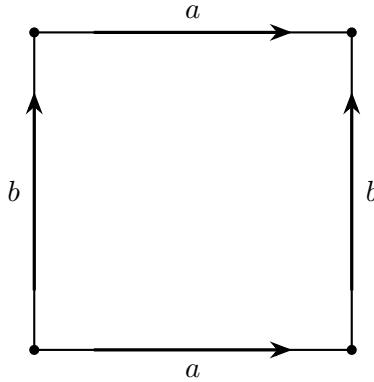


Figure 1: A square polygonal model for the torus. Opposite edges carrying the same label and arrow direction are identified.

After edge identifications, the four vertices become one point. Paired horizontal edges become one loop a , and paired vertical edges become one loop b . The result is a wedge of two circles if we don't include the interior of the square. Its fundamental group is the free group $F(a, b)$. If we attach the interior we can make the boundary path contractible which gives us the relation

$$aba^{-1}b^{-1} = 1.$$

This relation is equivalent to $ab = ba$, so every element can be written in the form $a^m b^n$.

Consequently,

$$\pi_1(T^2) \cong \langle a, b \mid aba^{-1}b^{-1} = 1 \rangle \cong \mathbb{Z} \oplus \mathbb{Z}.$$

3.2 The real projective plane

We can use a disk where every boundary point is identified with its antipodal point to model the real projective plane

$$x \sim -x \quad (x \in \partial D^2).$$

Choose antipodal points A and B . This divides the circumference into two semicircle pieces. Going from one endpoint to the other on a piece gives us an arc. Once we identify A and B , the image of each arc is a loop. The two arcs are themselves identified and become the same loop, denoted c .

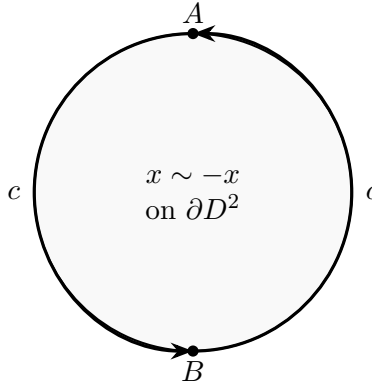


Figure 2: A disk model for $\mathbb{RP}^2$. The arrows show that a trip around the entire boundary traverses the same quotient loop c twice.

One traversal of the circumference, or boundary, of the disk becomes c^2 in the quotient. Since the boundary bounds the filled disk, this loop is contractible, so

$$\pi_1(\mathbb{RP}^2) \cong \langle c \mid c^2 = 1 \rangle \cong \mathbb{Z}/2\mathbb{Z}.$$

To provide some intuition for this, imagine you draw a perfect circle on a piece of paper. Cut off everything outside of the circle and everything inside the circle, leaving the marks you used to denote the circle alone. Take a ruler and define two points, A and B , directly across each other. Then cut the circle along those points. If you put those two arcs you just made on top of each other, they should look indistinguishable from each other. You can even tape the arcs to make two mini circles, and put them on top of each other. This gives a mental picture of what we mean when we say that going around one circle is the same as saying we went around the other. The actual interior of the disk is not folded or identified point-for-point.

4 Wedges of circles and attaching a two-cell

4.1 The fundamental group of a wedge of circles

Let

$$W_n = \bigvee_{i=1}^n S^1$$

be a wedge of n circles, meaning that all circles share one common basepoint.

Proposition 4.1. *For $n \geq 1$,*

$$\pi_1(W_n) \cong F_n,$$

the free group on n generators.

Proof. For $n = 1$, we assume that $\pi_1(S^1) \cong \mathbb{Z}$, which is the free group on one generator. Using induction, we write $W_n = W_{n-1} \vee S^1$. Choose open neighborhoods that deformation retract onto the two corresponding wedge summands and whose intersection is contractible. By the Seifert–van Kampen theorem,

$$\pi_1(W_n) \cong \pi_1(W_{n-1}) * \pi_1(S^1) \cong F_{n-1} * \mathbb{Z} \cong F_n.$$

□

Here $*$ denotes the free product of groups.

A geometric representation is that a loop in W_n is a word recording the particular circle traversed, the direction of its traversal, and the order of its traversal. The only automatic reductions are adjacent inverse cancellations because we have no two-dimensional region filling any of the loops.

4.2 The two-cell attachment lemma

Let $f : S^1 \rightarrow X$ be a loop in a path-connected space X . The notation

$$X \cup_f D^2$$

means that the boundary of a disk is glued to X according to f .

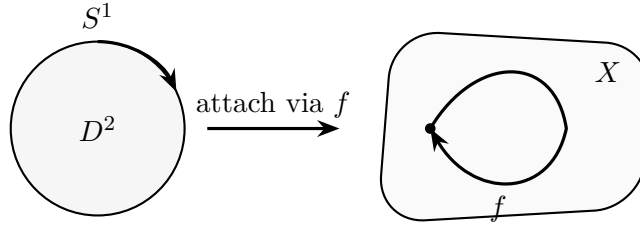


Figure 3: Attaching a two-cell fills the loop f with a disk.

Lemma 4.2 (Two-cell attachment lemma). *If X is path-connected and $f : S^1 \rightarrow X$ is a based loop, then*

$$\pi_1(X \cup_f D^2) \cong \pi_1(X) / \langle\langle [f] \rangle\rangle.$$

Proof. Let $Y = X \cup_f D^2$. Choose an open set $U \subseteq Y$ containing X and an annular collar of the attached disk. This set deformation retracts onto X , so $\pi_1(U) \cong \pi_1(X)$. Then, choose a second open set V around the center of the disk and enlarge it slightly into the collar. The set V is contractible. The intersection $U \cap V$ deformation retracts onto a circle.

A generator of $\pi_1(U \cap V) \cong \mathbb{Z}$ maps to $[f]$ in $\pi_1(U)$, because the annular loop can be pushed outward to the attaching boundary. The same generator maps to the identity in $\pi_1(V)$, because it bounds a disk there. Thus, Seifert–van Kampen identifies $[f]$ with the identity in $\pi_1(Y)$. Algebraically, the resulting group is the quotient of $\pi_1(X)$ by the normal closure of $[f]$. □

In summary we can notice the connection between transitioning from geometry to algebra. Geometrically the disk is filling the loop and algebraically the quotient is adding the relation $[f] = 1$.

5 Fundamental groups of closed surfaces

5.1 Orientable surfaces

A genus- g orientable surface has a $4g$ -gon model with boundary word

$$a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}.$$

Pairing all the edges allows the vertices to become one point and the glued boundary becomes a wedge of $2g$ circles. By theorem 4.1, its fundamental group is free on

$$a_1, b_1, \dots, a_g, b_g.$$

The polygon interior is a two-cell attached along the product of commutators

$$[a_1, b_1] \cdots [a_g, b_g].$$

Applying theorem 4.2 gives the following presentation.

Proposition 5.1. *For $g \geq 1$,*

$$\pi_1(\Sigma_g) \cong \langle a_1, b_1, \dots, a_g, b_g \mid [a_1, b_1] \cdots [a_g, b_g] = 1 \rangle.$$

Per handle, we notice there are two basic loop directions. The full boundary route around all the handles becomes contractible because it bounds the polygon's interior.

5.2 Nonorientable surfaces

A nonorientable surface N_k has a $2k$ -gon model with boundary word

$$c_1 c_1 c_2 c_2 \cdots c_k c_k = c_1^2 c_2^2 \cdots c_k^2.$$

The glued boundary is a wedge of k circles, and the polygon interior is a two-cell attached along the displayed word. Thus:

Proposition 5.2. *For $k \geq 1$,*

$$\pi_1(N_k) \cong \langle c_1, \dots, c_k \mid c_1^2 \cdots c_k^2 = 1 \rangle.$$

For $k = 1$, this recovers the projective plane $\mathbb{RP}^2$. For $k = 2$, it gives one standard presentation of the Klein bottle group.

6 Abelianization calculations

6.1 The orientable case

Commutators become the identity within the abelianization. Thus, the defining relation in theorem 5.1 imposes no remaining condition. The $2g$ generators are unrestricted, giving

$$\boxed{\pi_1(\Sigma_g)^{\text{ab}} \cong \mathbb{Z}^{2g}.}$$

For every $g \geq 1$, this group is nontrivial, so $\pi_1(\Sigma_g)$ is nontrivial.

6.2 The nonorientable case

After abelianization we switch to additive notation. The relation

$$c_1^2 \cdots c_k^2 = 1$$

becomes

$$2c_1 + \cdots + 2c_k = 0, \quad \text{or equivalently} \quad 2(c_1 + \cdots + c_k) = 0.$$

Set

$$d_1 = c_1 + \cdots + c_k, \quad d_i = c_i \quad (2 \leq i \leq k).$$

This is a valid change of generators because

$$c_1 = d_1 - d_2 - \cdots - d_k.$$

In the new coordinates, the only relation is $2d_1 = 0$. Hence, d_1 generates a copy of $\mathbb{Z}/2\mathbb{Z}$, while $d_2, \dots, d_k$ generate $k-1$ free integer directions. Therefore

$$\pi_1(N_k)^{\text{ab}} \cong \mathbb{Z}^{k-1} \oplus \mathbb{Z}/2\mathbb{Z}.$$

This is nontrivial for every $k \geq 1$, so $\pi_1(N_k)$ is nontrivial.

We can witness that after simplification in each case we have a nontrivial result. Regarding abelianization in the orientable case we see that two integer directions survive for each handle. In the nonorientable case there is a parity obstruction, the $\mathbb{Z}/2\mathbb{Z}$ term, that always survives.

7 Proof of the two-dimensional theorem

Note that S^2 is simply connected. First, cover it by two slightly enlarged hemispheres. Each is contractible and the intersection between them is an annular neighborhood of the equator. Seifert–van Kampen implies that the fundamental group of the union is trivial.

Proof of theorem 1.1. Let M be a closed, connected, simply connected surface. By theorem 2.1, M is homeomorphic to one of

$$S^2, \quad \Sigma_g \ (g \geq 1), \quad N_k \ (k \geq 1).$$

For $g \geq 1$,

$$\pi_1(\Sigma_g)^{\text{ab}} \cong \mathbb{Z}^{2g} \neq \{0\},$$

so $\pi_1(\Sigma_g)$ is nontrivial. For $k \geq 1$,

$$\pi_1(N_k)^{\text{ab}} \cong \mathbb{Z}^{k-1} \oplus \mathbb{Z}/2\mathbb{Z} \neq \{0\},$$

so $\pi_1(N_k)$ is also nontrivial. Since M is simply connected, neither case is possible. The only remaining possibility is

$$M \cong S^2.$$

□

8 Conclusion

We use the classification theorem to reduce the problem to a list of possible surfaces, but the fundamental-group calculations explain why all non-spherical surfaces must be eliminated. First, we use polygonal models to turn the geometry of edge identifications into generators and relations. Then we observe that attaching the polygon interior creates one relation, and finally abelianization retains the information we need to prove nontriviality.

Conceptually throughout this process, attaching the topological ideas to themes we are learning in our real analysis seminar made this project most approachable. For example, thinking about how $\mathbb{Z}/5\mathbb{Z}$ groups integers into equivalence classes using their remainders after division by 5 helped me understand how a similar structure works when using a quotient on a topological space. Remaining in the 2D space also helped me visualize much of the topological machinery that was going on as we deformed objects. Overall, I think this was an interesting introduction to Topology and I'm looking forward to exploring the subfield more.

References

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